

PEDIATRIC PROTOCOLS

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GENERAL PEDIATRIC CARE GUIDELINES

The key to quality pediatric care lies in the realization that children are not small adults. Scaled down equipment and smaller drug dosages are only the beginning. Pediatrics requires a different approach to patient care. The following guidelines should be kept in mind when treating pediatrics:

- The age range in pediatrics can make obtaining a history difficult but you should never dismiss the child's history.
- Initiate assessment of the pediatric patient using the PAT (Pediatric Assessment Triangle) to quickly evaluate appearance, work of breathing, and circulation status. A rapid cardiopulmonary assessment should be performed on all patients on initial contact and after each intervention.
- Cardiac arrest is seldom a sudden event. It is most often the results of a progressive deterioration of the circulatory (shock) and respiratory (hypoxia) systems.
- Hypoxia produces a reflex bradycardia in children. Any change in respiratory rate should be evaluated for a corresponding change in heart rate and vice-versa.
- Aggressive airway control and ventilation should be a priority, however in cardiac arrest, high quality CPR and early defibrillation should never be delayed.
- Intraosseous (IO) lines are the preferred route of access for all cardiac arrest victims. IO's may also be placed in a pediatric patient that is critically ill and needs immediate life-saving intervention. For example, such a patient may be displaying signs and symptoms of inadequate tissue perfusion (pale, cool, cyanotic or diaphoretic skin), altered level consciousness (lethargy), or profound hypotension defined as a systolic blood pressure less than $(70 + (\text{age in years} \times 2))$.
- Any medication directed to be given IV may also be given IO.
- Try to involve the parents as much as possible without compromising patient care.
- In a case of obvious death, CPR should be performed if it is the parent's wishes. Never leave the parent with the impression that something else could have been done.
- The cardiac monitor and other necessary equipment needed for resuscitation should be taken to the immediate side of any unconscious patient (or any patient presumably or likely to be in cardiac arrest until proven otherwise).
- Capnometry (waveform capnography) should be used in any critical pediatric patient to confirm placement of advanced airways and to monitor the perfusion, ventilation, and respiration status of the patient (This includes side stream or mainstream sampling). Sumner EMS carries pediatric side stream (nasal cannula) and mainstream ETCO2 monitoring equipment.
- Carbon Dioxide (CO) levels should be assessed on any patient presenting with altered mental status where there is potential or suspicion for CO exposure.
- Parents/guardians should be allowed to ride with the pediatric patient to the hospital and attempts should be made to not separate children from parents, unless the child is critical and it is best for everyone for the parent to ride elsewhere. (Also see COVID protocol)
- Temperature assessment for pediatric patients should be done on patients presenting with general sickness, febrile illness, seizures, hypothermia, or heat related illness.
- Sumner County EMS ambulances are only equipped with temperature monitoring equipment to perform oral thermometry or if available, the Zoll monitor temperature skin contact probe may also be adhered to the patient. At no time will EMS personnel assess rectal temperatures in the pre-hospital environment.
- Do not use an adult (normal size) finger lancet on an infant or small child's finger. This can lead to a laceration that will require treatment itself. Obtain samples from the heel in pediatric patients this small.

PEDIATRIC – EMERGENCY NEEDLE CRICOTHYROTOMY**OVERVIEW:**

A cricothyrotomy is a procedure to establish an emergency airway. It is an invasive procedure with multiple inherent complications and should be performed only on patients that are at high risk of death if an immediate airway is not established. One must first consider/attempt all alternative airway measures (e.g. OPA, NPA, ET intubation, supraglottic device, etc.). While every attempt should be made to transport to the closest emergency department for a more controlled setting, but no patient under the care of Sumner County EMS should die secondary to airway obstruction.

REQUIREMENTS:

- Be a licensed paramedic credentialed through the Deputy Chief of Training.
- Must have completed bi-annual training sessions as required by Sumner County EMS.
- No longer required to contact on-line medical control, this is a standing order now.

INDICATIONS:

- Emergent Need Only.
- **Inability to intubate and inability to ventilate.**
- Indicated for extremis presentations of:
 - Foreign body airway obstruction.
 - Airway burns.
 - Anatomical injury
 - Anaphylaxis unresolved by Epi
 - Epiglottitis/croup unresolved by Epi

COMPLICATIONS:

- Bleeding.
- Misplacement (esophageal or soft-tissue placement).
- Damage to surrounding structures such as vocal cords, esophageal or tracheal damage.
- Infection.

CONTRAINDICATIONS:

Given that you will only be performing this procedure on patients who have a very high probability of dying without it, most contra-indications would therefore be relative. The following are examples of situations that may prohibit needle cricothyrotomy:

- Age > 10 years old / Adult sized, refer to the adult cricothyrotomy protocol if needed.
- Able to ventilate with less invasive therapies.
- Any causes preventing identification of appropriate anatomical landmarks.

EQUIPMENT NEEDED:

• Non latex gloves	• ½ inch medical tape
• Sharps container	• 3.0 ET tube (connector)
• Oxygen Supply BVM	• NS Flush
• ChloroPrep/antiseptic	• Neo-T Resuscitator Device
• 14G IV Angiocath	• 90-degree IV extension set

PEDIATRIC – EMERGENCY NEEDLE CRICOTHYROTOMY (Procedure)

EMR

- Obtain basic assessment
- Attempt to open airway manually with head-tilt chin-lift or jaw thrust in trauma patients
- If airway obstruction is visible and able to be reached, remove obstruction manually (No blind sweep)
- Attempt to ventilate with BVM

EMT

- Obtain vital signs including SpO₂, blood pressure, and ECG attachment
- Attempt to establish a less invasive airway (Supraglottic device)

AEMT

- Prepare for waveform capnography monitoring and apply to ET Tube after placement
- Establish vascular access as indicated

PARAMEDIC

- Position patient supine and extend the neck as needed to improve anatomic view
- Prep neck with antiseptic
- Insert 14ga angiocath at a 45-degree angle with hub held caudally (toward the feet)
- Withdraw air during insertion through NS flush that is half filled with fluid
- Assess for penetration into trachea by watching for air bubbles in syringe
- Careful not to insert needle too far, the diameter of the pediatric trachea is similar in size to the child's pinky finger
- Advance the catheter flush to the skin, remove needle and dispose of properly
- Attach IV extension set directly to the 14ga angiocath hub (**DO NOT** use the IO extension set)
- Attach 3.0 mm ET tube ventilation adapter to the end of the IV extension set
- Attach Neo-T style resuscitator or pediatric BVM to ET tube ventilation adapter and initiate ventilations
- If Neo-T resuscitator is used, ensure that the Neo-T is set to "HIGH" flow rate (RED)
- The O₂ source powering the Neo-T resuscitator should be running at high flow (15 Lpm+)
- Adjust built-in PEEP dial titrated to effect to ensure effective exhalation
- If pediatric BVM is used, ensure that the "Override" clamp is in place
- Confirm that chest rise and fall are present
- Secure as soon as possible with tape provided. This should be treated as fragile as a pediatric/infant intubation and the cannula should not be released until it can be secured.

PEDIATRIC FACILITATED INTUBATION AND RSI

OVERVIEW:

Rapid Sequence Intubation (RSI) is a series of maneuvers utilizing sedation and paralysis to establish an advanced airway in a critically ill patient. This is an advanced procedure with a potential for high-risk complications and should only be performed as an absolute life-saving procedure. It should only be performed after all other less invasive forms of airway control have been attempted or considered. At no time should a paramedic feel pressured to perform this procedure if he or she is not comfortable with its application on a given patient. ***Updated 2019 - All paramedics may initiate this procedure with the administration of choice sedatives (chemically facilitated intubation). However, you must meet the requirements below in order to perform full RSI, to include administration of paralytics.***

REQUIREMENTS TO PERFORM RSI:

- Be a licensed paramedic and employee of Sumner County EMS for at least 6 months, or experienced personnel may be approved by the Medical Director.
- Be in good standing with the service regarding clinical issues.
- Complete bi-annual airway, RSI and cricothyrotomy training courses.

INDICATIONS:

To establish an airway in a patient who is at risk of death secondary to loss of airway or inability to ventilate, and the airway cannot be controlled by conventional means. Examples of patients in which pre-hospital RSI might be indicated include, but are not limited to the following:

- Facial or head trauma patients with loss of airway control.
- Severe respiratory distress with hypoxia and/or respiratory exhaustion.
- Burn patients with airway involvement and respiratory distress.
- Overdose with loss of airway protection and hypoxia.

CONTRAINDICATIONS:

- Allergy to any one of the agents.

CONTRAINDICATIONS TO SUCCINYLCHOLINE:

- History of malignant hyperthermia.
- Renal failure.
- Spinal cord injury greater than 24 hours old or neuromuscular disease.
- Severe burns greater than 8 hours old.
- Massive crush injuries.
- Pesticide poisoning.
- Penetrating eye injuries.

FACILITATED INTUBATION AND RSI (PROCEDURE) – (Continued)

EMR

- Position patient and open airway manually with head-tilt chin-lift or jaw thrust maneuver
- Insert OPA as indicated
- Preoxygenate by administering 100% oxygen and ventilate with BVM
- Obtain basic assessment including manual blood pressure

EMT

- Obtain vital signs including SpO2
- Place electrodes on patient to obtain ECG
- Attempt less invasive airway device (NPA, OPA, or Supraglottic Device)
- Place ETCO2 NC on patient at 15 LPM to begin hyperoxygenation

AEMT

- Begin capnography monitoring using ETCO2 NC
 - Note that ETCO2 measurement may be inaccurate while using high-flow oxygen >5 LPM
- Obtain vascular access
- If hypotensive, begin fluid resuscitation as indicated

PARAMEDIC

- **In children less than 8 years old, consider pretreating with 0.02 mg/kg of Atropine IV/IO.**
- Give **SEDATIVE** (Induction) – Use appropriate, available induction agent:
 - Ketamine 1-2 mg/kg IV/IO
 - First choice induction agent (sedative)
 - Avoid if patient is hypertensive, or in presence of tachyarrhythmias or acute MI
 - Etomidate 0.25 mg/kg IV/IO, max of 20 mg, when Ketamine is contraindicated (**first alternative to Ketamine**).
 - Versed 0.2 mg IV/IO, max of 5 mg (**only used as an alternative to Etomidate**).

***** CONSIDER ATTEMPTING FACILITATED INTUBATION AFTER AVAILABLE INDUCTION AGENT *****

****** REFER TO CONTRAINDICATIONS FOR SUCCINYLCHOLINE ON PREVIOUS PAGE PRIOR TO GIVING! ******

- Give PARALYTIC (short-acting)
 - Administer Succinylcholine 1.5 mg/kg IV/IO, maximum dose of 100 mg, or Rocuronium 1 mg/kg IV/IO, maximum dose of 100 mg.

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FACILITATED INTUBATION AND RSI (Continued) – POST-INTUBATION MANAGEMENT

PARAMEDIC

- Placement and Proof
 - Intubate when patient becomes flaccid, often after fasciculations. If the patient cannot be intubated after 2 attempts, then use an alternative airway such as the iGel or basic airway adjuncts and continue to ventilate patient until the Succinylcholine wears off
 - Confirm placement with capnography / capnometry, auscultation, etc.
 - **(Minimum of 5 verification techniques)**

*For any patient who is already intubated (interfacility transfers), **ALL PARAMEDICS** can begin with this section to maintain sedation, go to long-acting paralytics as needed, see below...*

- Post-Intubation Management
 - Administer 100% Oxygen, and titrate to > 94% when possible
 - Maintain ETCO₂ of 35-45 mmHg when possible
 - Secure ET Tube
 - Avoid delivering excessive volumes / pressures with the BVM to prevent risk of barotrauma
 - Use a PEEP valve (5-10 cmH₂O) in patients with a stable blood pressure
- Give maintenance **SEDATION** and **PAIN CONTROL**. This **MUST BE DONE!**
 - Ketamine 0.5-1 mg/kg IV/IO, one time dose, if Ketamine was used for induction agent prior to RSI
 - Versed 0.1-0.2 mg/kg IV/IO, max of 5 mg
 - Repeat as necessary and titrate to desired effect
 - Fentanyl 1 mcg/kg IV/IO, max of 100 mcg, may also be given
 - Repeat as necessary and titrate to desired effect
- Administer long-acting paralytic after correct placement is assured
 - Rocuronium (Zemuron) 1 mg/kg IV/IO, max of 100 mg
 - Actual dose range is 0.6 - 1 mg/kg. May give 1 mg/kg for ease of administration.
 - Norcuron (Vecuronium) 0.1 mg/kg IV/IO, max of 10 mg, if Rocuronium is unavailable.
- Transport without delay

GUIDELINES FOR CUFFED ETT UTILIZATION IN THE PEDIATRIC PATIENT

(Adopted from Tennessee EMS for Children's Education Guidelines)

When using a cuffed ETT for a pediatric patient:

1. Select the appropriate size ETT (typically ½ size smaller than the recommended uncuffed ETT size as per length-based resuscitation tape).
2. Check the integrity of the cuff prior to insertion.
3. After insertion of the ETT, inflate the cuff as necessary to achieve minimal air leak around the ETT (amount of air not to exceed the manufacturer's specification for maximum air inflation).
4. Completely deflate the cuff prior to removal of the ETT.

It is accepted practice to use cuffed ETTs in any patient. However, providers must understand that they may not need to inflate the cuff in all cases, especially in a neonate patient.

When you are inflating a pediatric cuffed ETT, only inflate the cuff with enough air to inflate the pilot cuff, do not over-inflate. This may take less than the normal 10 ml of air in an adult ETT cuff.

**References:**

Hoffman RJ, Dahlen JR, Lipovic D, Stürmann KM. Linear Correlation of Endotracheal Tube Cuff Pressure and Volume. Western Journal of Emergency Medicine. 2009;10(3):137-139.

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<http://www.joacp.org/text.asp?2011/27/3/358/83682>. Lichtenthal, PL and Borg, UB. Endotracheal cuff pressure: role of tracheal size and cuff volume. Critical Care. 2011;15(1):147. <http://ccforum.com/content/15/S1/P147>.

OROGASTRIC (OG) TUBE INSERTION PROTOCOL

PARAMEDIC

Paramedics may use this protocol to facilitate gastric decompression of the intubated patient after endotracheal intubation has been performed and verified, or with the use of the iGel Airway.

Contraindications:

- Known esophageal disease (such as varices).
- Gastric Bypass.
- Penetrating neck trauma.

Especially consider use in:

- Submersion injuries (near drowning).
- Intubated pediatric patients.
- Any patient who has been aggressively artificially ventilated.

Equipment:

1. 8 or 10 French Sump Catheter.
2. Lubricant (water-soluble).
3. 60 mL catheter tip syringe.
4. Tape.

Procedure:

1. Estimate the length of the tube needed to reach the stomach by measuring the tube from the corner of the mouth to the earlobe and down to the xiphoid process. Mark the length with tape.
2. Lubricate the Salem sump tube with a water-soluble lubricant (e.g. KY Jelly).
3. Insert the tube through the oropharynx OR through the gastric access lumen on the iGel Airway until the marked depth is reached.
4. If the tube coils in the posterior pharynx, direct laryngoscopy can be utilized to place the tube in the esophagus. If the patient begins to cough, you may be advancing into the trachea. Reattempt insertion...
5. Verify placement (see OG Placement Verification).

Placement Verification:

1. Using a 60 mL catheter tip syringe, instill 10 mL (consider 2-5 mL in children less than one year of age) of air into tube and auscultate over epigastrium for air sounds.
2. Aspirate for gastric contents and assess for cloudy, green, tan, brown, bloody or off-white color contents consistent with gastric contents.
3. Secure tube with tape.

Gastric Decompression:

1. Attach the tube to suction (and suction intermittently as needed). You can also provide continual suction by setting the suction device at 20-60 mmHg negative pressure (TORR).
2. If suction is not readily available, connect the 60 mL syringe to the tube with keeping the (blue) air vent patent. This will allow the sump function of the tube to work until suction can be applied and will also prevent gastric contents from leaking from the tube.

Special Note:

- If you cannot place the OG tube quickly (no more than 2 attempts), forego the procedure – DO NOT delay transport.
- The blue air vent must remain patent to ensure proper sump function and prevent damage to the gastric lining during suction.

PEDIATRIC PAIN MANAGEMENT (TRAUMATIC)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of traumatic pain.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- If there is an obvious fracture, refer to "Fractures General Care" protocol.

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading

PARAMEDIC

- For acute traumatic injuries causing extreme pain admin:
 - Fentanyl – 1 mcg/kg slow IV/IO
 - If allergic to fentanyl admin:
 - Morphine 0.05 mg/kg IV/IO, Max initial dose of 5 mg
- If no IV access available or patient only has minor injuries admin:
 - Fentanyl 2 mcg/kg IN or IM
 - If allergic to Fentanyl admin:
 - Morphine 0.1 mg/kg IM
- Fentanyl may be repeated, titrated to effect, given 5-10 min apart:
 - 0.5 mcg/kg IV/IO
 - 1 mcg/kg IN or IM

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PEDIATRIC PAIN MANAGEMENT (TRAUMATIC) – (Continued)

- Morphine may be repeated, titrated to effect given 5 min after initial:
 - 0.05 mg/kg IV/IO – Max of 5 mg
 - 0.1 mg/kg IM
- Ketamine to assist with Pain Management is NOT to be given to pediatric patients without physician orders.
- Use EXTREME caution when administering pain meds to pediatric patients, use weight-based charts to confirm dosages.
- **There are no standing orders for pain meds in Abdominal Pain or Medical Pain complaints.**

POSITIVE PRESSURE AIRWAY PROTOCOL, PART 1 – PEDIATRIC CPAP

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

AEMT

PARAMEDIC

CPAP = **Continuous Positive Airway Pressure**. CPAP delivers "positive pressure" to patients who have effective ventilation, but have ineffective respiration (exchange of O₂ and CO₂). Dyspnea patients who present with an altered mental status have likely progressed to respiratory failure. A patient with an altered level of consciousness is not a candidate for CPAP and needs more evasive treatments such as a BVM.

PEDIATRIC Settings:

(CPAP or PEEP): 2-5 cmH₂O, not to exceed 5 cmH₂O unless otherwise directed by a physician / on-line medical control. Ranges exceeding 10 cmH₂O should be avoided. For adolescents similar in size of an adult, adult ranges may be considered but should be approved by a physician prior to use. -- Attach the 2-5 cmH₂O valves to the CPAP device as needed. --

CPAP Indications:

For conscious and alert patients 12 years old or greater with moderate to severe respiratory distress (accessory muscle use or tripod position) from an underlying pathology or acute condition that affects respiration at the alveolar level.

- Bronchiolitis, Pneumonia or Asthma.
- Near drowning / submersion injuries.
- Pulmonary edema (see notes below):
 - May be seen as an underlying pathology in pediatric patients with special needs conditions.
 - May be seen as an acute presentation where there has been an inhalation injury from toxic gases such as chemical exposure or smoke inhalation on fire scenes.

CPAP Contraindications:

- Respiratory Arrest (needs BVM instead).
- Signs and symptoms of a pneumothorax or chest trauma.
- Active gastrointestinal bleeding or vomiting.
- Patient unable to follow verbal commands or altered mental status.
- Inability to properly fit the CPAP mask and strap or tracheotomy.
- Overdoses.

CPAP Procedure:

1. Assure ABC's are intact, stabilize as needed, monitor ECG, SPO₂ and ETCO₂.
2. Set CPAP device to desired pressure setting (2-5 cmH₂O for pediatrics), titrate as needed.
3. Talk to the patient and/or parents, and explain the procedure / therapy.
4. Once the device is running, then apply to the patient. Use an appropriate size mask for the patient and ensure a good seal as best possible.
5. Constantly re-assess the patient, if their mental status deteriorates, discontinue and ventilate PRN. Nebulized medications such as Albuterol can be interlinked if necessary.

**Monitoring CPAP Effectiveness:**

Pediatric CPAP monitoring should include cardiac monitoring, ETCO₂, pulse oximetry, and frequent assessments of lung sounds, monitor for worsening gastric distension and temperature assessment.

- Improving skin color, mental status.
- Improving respiratory [tidal volume, lung sounds.
- Decreasing respiratory rate, accessory muscle use and retractions.
- Decreasing anxiety or agitation.
- A normalizing heart rate.

POSITIVE PRESSURE AIRWAY PROTOCOL, PART 2 – PEDIATRIC PEEP

*Positive End Expiratory Pressure use in **Pediatric** Patients*

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

AEMT

PARAMEDIC

PEEP = **Positive End Expiratory Pressure**. PEEP delivers "positive pressure" at the end of exhalation to **prolong the exhalation time, helping to exhale CO₂**. This is delivered by attaching a PEEP valve onto a **BVM** for patients in respiratory failure or arrest who require ventilatory support. *PEEP may be beneficial; however, it can worsen hypoperfusion, therefore it should NOT be used in this situation or cardiac arrest.*

PEDIATRIC Settings:

(CPAP or PEEP): 2-5 cmH₂O, not to exceed 5 cmH₂O unless otherwise directed by a physician / on-line medical control. Ranges exceeding 10 cmH₂O should be avoided. For adolescents similar in size of an adult, adult ranges may be considered but should be approved by a physician prior to use. -- Attach the 2-5 cmH₂O valves to the CPAP device as needed. --

PEEP Indications:

Most patients who require artificial ventilation will benefit from PEEP. Physiologic PEEP exists in people upon natural exhalation. PEEP valve use should especially be considered in patients who require BVM ventilation and have underlying pathology that is affecting respiration, such as pulmonary edema, COPD, Asthma, pneumonia, atelectasis, or near drowning patients.

PEEP Contraindications:

- Signs and symptoms of a pneumothorax or chest trauma.
- Tracheotomy (precaution).
- Active gastrointestinal bleeding or vomiting.
- Avoid use in hypotension or cardiac arrest.

PEEP Procedure:

- Attach PEEP valve onto exhalation port of BVM and adjust settings by rotating red knob.



PEDIATRIC – REFUSALS (REQUIREMENTS TO BE MET)

PARAMEDIC ONLY

In situations where there is considerable mechanism of injury or possibility for medical emergencies with pediatric patients, refusals of transport should be avoided if at all possible.

However, it is recognized that parents or designated caregivers/guardians may elect to transport patients on their own, or even forego further assessment, treatment, or transport. Situations where refusal of transport should be avoided whenever possible are as follows:

- BRUE (Brief Resolved Unexplained Event) / apparent life-threatening event.
- Anaphylaxis / allergic reaction / envenomation.
- Near drowning.
- Altered Mental Status.
- Seizure.
- Possible head injuries.
- Dyspnea / Breathing difficulty.
- Medication overdose or poisoning (including chemical exposure).
- **Any suspected abuse or neglect situation.**

In order for caregivers/guardians to consider refusing EMS transport, EMS providers must be able to determine the following:

- The patient is **alert and oriented** appropriate for their age.
- The patient has **effective** work of **breathing**.
- The patient is hemodynamically **stable**.
- The patient will be left in a **safe** environment.
- The patient will be in the **care** of an appropriate guardian/caregiver.

When possible, a complete set of vital signs shall be obtained prior to making any decision not to transport a pediatric patient. This may also include other diagnostics as appropriate for patient's condition. (Blood glucose, ECG, pulse oximetry, etc.)

If any parent, guardian, or otherwise deemed caregiver is refusing to allow transport against medical advice of EMS personnel, proceed as follows:

- Contact on-duty supervisors immediately.
- Avoid conflict, when possible, involve law enforcement as needed.

PEDIATRIC PATIENT SAFE TRANSPORT GUIDELINES

EMR

EMT

AEMT

PARAMEDIC



The Do's and Don'ts of Transporting Children in an Ambulance

Approximately 6 million children are transported by emergency medical services (EMS) vehicles each year in the United States. There are risks of injury associated with transport that can be minimized. An ambulance is NOT a standard passenger vehicle. Unlike the well-developed and publicized child passenger safety standards and guidelines, specifications for the safe transport of ill and injured children in ambulances are still under development. Standard automotive safety practices and techniques cannot be applied directly to EMS vehicle environments due to biomechanical and practical differences. Caution is encouraged in the application of passenger vehicle principles to ambulances and in the utilization of new and unproven products.

The Emergency Medical Services for Children (EMSC) Program supports efforts to improve the safety of pediatric patients being transported in EMS vehicles. Through an EMSC grant, the Division of Pediatric Emergency Medicine at Johns Hopkins Children's Center conducted the first ever ambulance vehicle crash safety tests using instrumented crash test dummies to test the performance of these vehicles and a variety of occupant and equipment restraint practices currently available or in use. The data from these and other detailed safety tests are currently being analyzed and will be an integral element in the development of safety guidelines for the transport of ill and injured children.

Currently, the National Highway Traffic Safety Administration's (NHTSA) Office of EMS and Occupant Protection Division are working with the EMSC Program to develop recommendations for how EMS can safely transport ill or injured – and uninjured – children in ambulances.

Until consensus recommendations are available, certain practices can significantly decrease the likelihood of a crash, and in the event of a crash or near collision, can significantly decrease the potential for injury. These practices are listed below in the format of "Do's" and "Don'ts."

Importantly, as is mandated in several states, the NHTSA Emergency Vehicle Operating Course (EVOC) National Standard Curriculum, or its equivalent, is an integral part of this transport safety enhancement.

Do's

- ☒ DO drive cautiously at safe speeds observing traffic laws.
- ☒ DO tightly secure all monitoring devices and other equipment.
- ☒ DO ensure available restraint systems are used by EMTs and other occupants, including the patient.
- ☒ DO transport children who are not patients, properly restrained, in an alternate passenger vehicle, whenever possible.
- ☒ DO encourage utilization of the EVOC, National Standard Curriculum.

Don'ts

- ☒ DO NOT drive at unsafe high speeds with rapid acceleration, decelerations, and turns.
- ☒ DO NOT leave monitoring devices and other equipment unsecured in moving EMS vehicles.
- ☒ DO NOT allow parents, caregivers, EMTs or other passengers to be unrestrained during transport.
- ☒ DO NOT have the child/infant held in the parent, caregiver, or EMT's arms or lap during transport.
- ☒ DO NOT allow emergency vehicles to be operated by persons who have not completed EVOC or an equivalent course.

FAMILY VIOLENCE / CHILD ABUSE PROTOCOL

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of family violence or child abuse.

EMR	EMT	AEMT	PARAMEDIC
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INDICATORS:

- Fear of household member.
- Reluctance to respond when questioned.
- Unusual isolation, unhealthy, unsafe living environment.
- Poor personal hygiene/inappropriate clothing.
- Conflicting accounts of the incident.
- History inconsistent with injury or illness.
- Indifferent or angry household member.
- Household member refused to permit transport.
- Household member prevents patient from interacting openly or privately.
- Concern about minor issues but not major ones.
- Household with previous violence.
- Unexplained delay in seeking treatment.

Direct questions to ask when alone with patient and time available:

1. Has anyone at home ever hurt you?
2. Has anyone at home touched you without your consent?
3. Has anyone threatened you?
4. Are you afraid of anyone at home?

Signs and Symptoms:

- Injury to soft tissue areas that are normally protected.
- Bruise or burn in the shape of an object.
- Bite marks.
- Rib fracture in the absence of major trauma.
- Multiple bruising in various stages of healing.

Protocol:

1. Patient care is first priority.
2. If possible, remove patient from situation and transport.
3. Summons police assistance as needed.
4. If suspected sexual assault, then follow sexual assault protocol.
5. Obtain information from patient and caregiver.
6. Do not judge.
7. Report suspected abuse to hospital after arrival. Make verbal and written report.

Report Suspected Child Abuse and Neglect

State of Tennessee Phone

Number: 1-877-237-0004

Secure On-line site:

[hNps://apps.tn.gov/carat/](https://apps.tn.gov/carat/)

National Domestic Violence

Hotline: 1-200-799-SAFE (7233)

DEATH OF A CHILD AND SUDDEN INFANT DEATH SYNDROME (SIDS)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with an unexplained death of an infant.

EMR**EMT****AEMT**

There is no normal parental reaction to the death of a child or a SIDS event. Individual responses may range from emotional outbursts to apparent withdrawal. Rescuers should not make any assumptions or judgments. Maintain a professional demeanor at all times. Perform the initial assessment, environmental assessment, and focused history as part of the clinical process. Observe, assess, and document accurately and objectively.

1. Ensure scene safety.
2. Perform a scene survey to assess environmental conditions and mechanism of illness or injury.
3. Form a general impression of the patient's condition.
4. Observe standard precautions.
5. Establish patient responsiveness.
6. Assess airway and breathing. Confirm apnea.
7. Assess circulation and perfusion.

PARAMEDIC

1. Initiate cardiac monitoring. Confirm absent pulse.
2. Determine whether to perform further resuscitation measures: If patient does not exhibit lividity or rigor, proceed with cardiopulmonary resuscitation. During resuscitation, perform steps 11 and 12 below. Initiate transport. If patient exhibits lividity and rigor, do not resuscitate as permitted by medical direction. Proceed with step 10. Note: Lividity can be mistaken for bruising and evidence of abuse. Be careful not to make any assumptions or judgments.
3. Provide supportive measures for parents and siblings:
 - a. Explain the resuscitation process, transport decision, and further actions to be taken by hospital personnel or the medical examiner.
 - b. Reassure parents that there was nothing they could have done to prevent death.
 - c. Allow the parents to see the child and say goodbye.
 - d. Maintain a supportive, professional attitude no matter how the parents react.
4. Whenever possible, be responsive to parental requests. Be sensitive to ethnic and religious needs and make allowances for them.

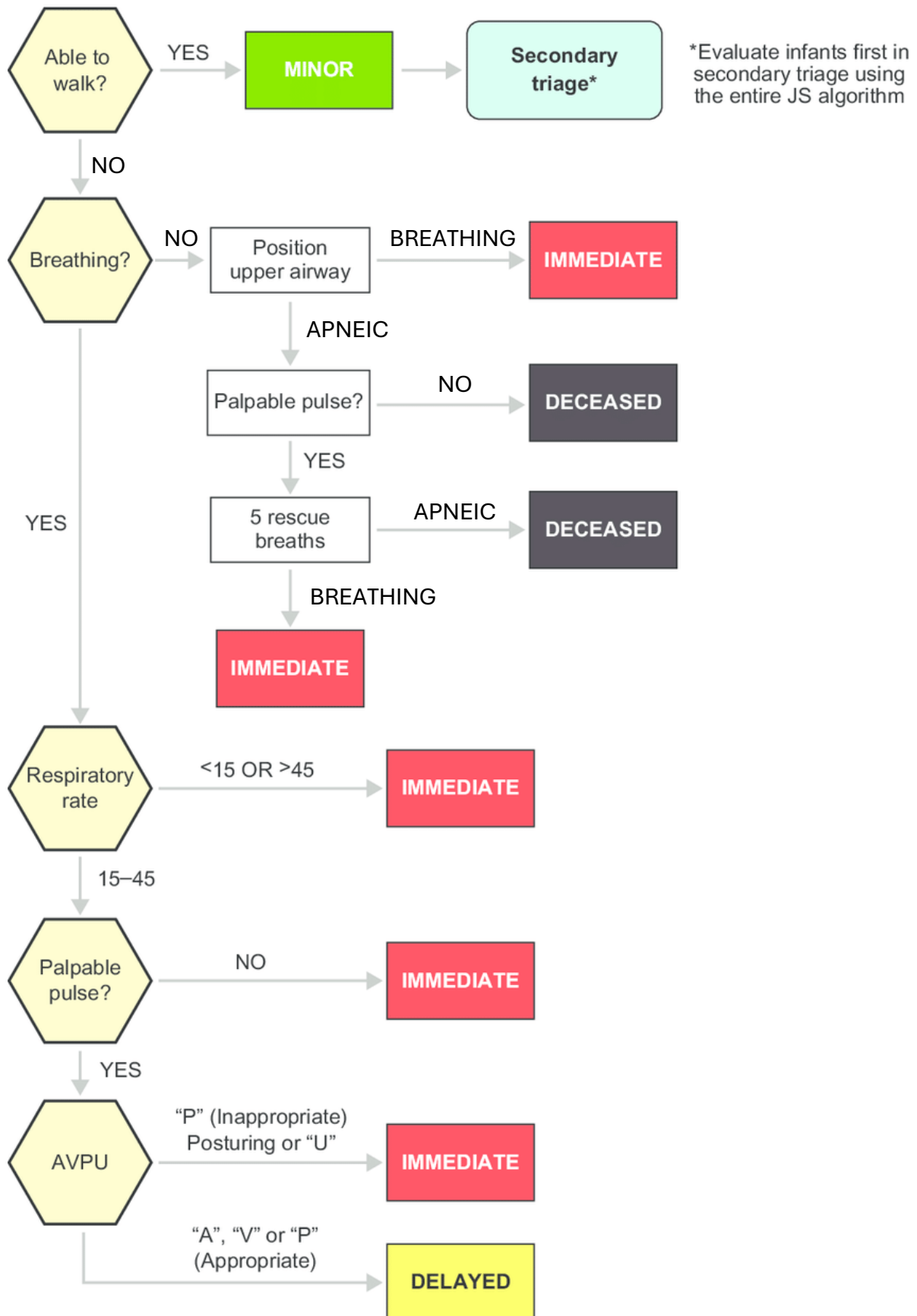
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DEATH OF A CHILD AND SUDDEN INFANT DEATH SYNDROME (SIDS) (Continued)

5. Obtain patient history using a nonjudgmental approach. Ask open-ended questions as follows:
 - a. Has the child been sick?
 - b. Can you describe what happened?
 - c. Who found the child? Where?
 - d. What actions were taken after the child was discovered?
 - e. Has the child been moved?
 - f. When was the child last seen before this occurred, and by whom?
 - g. How did the child seem when last seen?
 - h. When was the last feeding provided?
6. Reassess the environment. Document findings, noting the following:
 - a. Where the child was located upon arrival?
 - b. Description of objects located near the child upon arrival?
7. Unusual environmental conditions, such as a high temperature in the room, abnormal odors, or other significant findings?
8. If the parents interfere with treatment or attempt to alter the scene, initiate the following actions:
 - a. Remain supportive, sympathetic, and professional.
 - b. Avoid arguing with the parents or exhibiting anger.
 - c. Do not restrain the parents or request that they be restrained unless scene safety is clearly threatened.
9. Document the emergency call, including the following information:
 - a. Time of arrival.
 - b. Initial assessment findings and basis for resuscitation decision.
 - c. Time of resuscitation decision.
 - d. Time of arrival at hospital if resuscitation and transport were initiated.
 - e. Parental support measures provided if resuscitation was not initiated.
 - f. History obtained (note who provided the information).
 - g. Environmental conditions.
 - h. Time law enforcement personnel arrived on scene.
 - i. Time that scene responsibility was turned over to law enforcement personnel.

The priority of emergency medical services personnel on scenes involving infant/pediatric death is to provide expeditious transport and deliver emergency medical treatments. It is the primary role of law enforcement and medical examiners to perform detailed assessments of the scene, environment, and events surrounding the incident.

EMS personnel are not expected to perform any other duties from delivering assessment, treatment, and transport. Any additional information acquired during these phases of the call shall be documented appropriately, otherwise EMS shall not delay treatment or transport to gather information about the scene.

PEDIATRIC MCI TRIAGE - JumpSTART**JumpSTART pediatric MCI triage**

PEDIATRIC VENTRICULAR FIBRILLATION/PULSELESS V-TACH

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of ventricular fibrillation or pulseless ventricular tachycardia.

AEMT level may give Epinephrine 1:10,000 1 mg IV or IO in cardiac arrest situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Begin high-quality CPR
 - Compressions at 100-120/min
 - Compress 1/3 to 1/2 depth of the chest wall
 - Allow adequate chest recoil
 - Minimize interruptions
- Place AED and shock if advised
- Obtain vital signs as indicated

EMT

- Pulse oximetry
- Place oral airway and ventilate using a BVM with 100% O₂
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish Vascular access (IV/IO)
- Place supraglottic airway if:
 - Patient cannot be effectively ventilated with OPA and BVM
 - Paramedic is not available to direct advanced airway procedures
- Obtain EtCO₂ reading

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PEDIATRIC VENTRICULAR FIBRILLATION/PULSELESS V-TACH (Continued)

PARAMEDIC

- Defibrillate a 2 Joules/kg ASAP (Joule setting can be child's weight in pounds)
 - Repeat defib every 2 min AS NEEDED at 4 J/kg
- Secure patients' airway with most definitive adjunct available
- Administer Epinephrine 0.01 mg/kg IV/IO (1:10,000)
 - Repeat every 3-5 min until ROSC or Termination of Resuscitation
 - No max dose
- Administer Amiodarone 5 mg/kg IV/IO bolus
 - Repeat ONCE only in 3-5 min at same dose if patient is still in shockable rhythm
 - If Amiodarone is unavailable, Lidocaine 1 mg/kg bolus IV/IO per dose, to a maximum of 3 mg/kg
- Consider H's and T's
 - Consider Mag Sulfate for Torsades de Pointes
 - Consider Narcan 0.1 mg/kg IV/IO or 0.2 mg/kg IM/IN

PEDIATRIC ASYSTOLE / PULSELESS ELECTRICAL ACTIVITY

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of asystole or pulseless electrical activity (PEA).

AEMT level may give Epinephrine 1:10,000 1 mg IV or IO in cardiac arrest situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Begin high-quality CPR
 - Compressions at 100-120/min
 - Compress 1/3 to 1/2 depth of the chest wall
 - Allow adequate chest recoil
 - Minimize interruptions
- Place AED and shock if advised
- Obtain vital signs as indicated

EMT

- Pulse oximetry
- Place oral airway and ventilate using a BVM with 100% O₂
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish Vascular access (IV/IO)
- Place supraglottic airway if:
 - Patient cannot be effectively ventilated with OPA and BVM
 - Paramedic is not available to direct advanced airway procedures
- Obtain EtCO₂ reading

CONTINUED ON NEXT PAGE

PEDIATRIC ASYSTOLE / PULSELESS ELECTRICAL ACTIVITY (Continued)

PARAMEDIC

- Secure patients' airway with most definitive adjunct available
- Administer Epinephrine 0.01 mg/kg IV/IO (1:10,000)
 - Repeat every 3-5 min until ROSC or Termination of Resuscitation
 - No max dose
- Consider H's and T's
 - Consider Mag Sulfate for Torsades de Pointes
 - Consider Narcan 0.1 mg/kg IV/IO or 0.2 mg/kg IM/IN

PEDIATRIC SYMPTOMATIC BRADYCARDIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of symptomatic bradycardia.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs as indicated

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish Vascular access (IV/IO)
- Obtain EtCO₂ reading

PARAMEDIC

- Acquire 12-lead initially and repeat 12 lead every 10 minutes
- Stable patients:
 - Administer Epinephrine 0.01 mg/kg 1:10,000 IV/IO
 - Repeat every 3-5 min as needed
 - Administer 0.1 mg 1:1,000 via ET tube if IV or IO is not available
 - If no response to O₂ or Epi, then admin:
 - Atropine 0.02 mg/kg IV
 - Minimum dose of 0.1 mg
 - May repeat ONCE in 3-5 minutes

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PEDIATRIC SYMPTOMATIC BRADYCARDIA (Continued)

- Unstable patients (Bradycardia with critical hypotension, altered mental status, unresponsive, or ischemic chest discomfort)
 - Initiate CPR ONLY IF:
 - Infant with HR LESS THAN 80 bpm
 - Child with HR LESS THAN 60 bpm
 - Follow “stable patient” flow for medication regimen
- Consider Organophosphate Poisoning as cause of bradycardia
 - If suspected admin:
 - Atropine 0.05 mg/kg (usual dose 1-5mg)
 - Repeat if needed in 5-15 min

PEDIATRIC WIDE COMPLEX TACHYCARDIAC (V-TACH)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of wide complex ventricular tachycardia with a pulse.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs as indicated

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish Vascular access (IV/IO)
- Obtain EtCO₂ reading

PARAMEDIC

- Acquire 12-lead initially and repeat 12 lead every 10 minutes
- Stable patients:
 - Administer Amiodarone 5 mg/kg IV drip slow over 20 minutes
 - Max 150 mg
 - If Amiodarone is unavailable, Lidocaine 1 mg/kg can be given, may repeat in 5-10 minutes at 0.5 mg/kg
- Unstable patients (Mottled skin/ Cyanosis, critical hypotension, altered mental status, unresponsive, or ischemic chest discomfort)
 - If the patient is conscious / responsive to pain, give:
 - Versed 0.1 – 0.2 mg/kg IV/IO
 - Consider Versed 0.2 mg IM if no vascular access is available
 - Max dose 2 mg IV or 5 mg IM

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PEDIATRIC WIDE COMPLEX TACHYCARDIA (V-TACH) (CONTINUED)

- Synchronized Cardioversion
 - Place monitor into synchronized mode and shock in the following sequence until patient converts:
 - 0.5 Joules/kg, then:
 - 1 Joules/kg, then:
 - 2 Joules/kg (Max dose), repeat at 2 J/kg as needed

PEDIATRIC TACHYCARDIA – NARROW WITH A PULSE

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of narrow tachycardia with a pulse.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs as indicated

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish Vascular access (IV/IO)
- Obtain EtCO₂ reading

PARAMEDIC

- Acquire 12-lead initially and repeat 12 lead every 10 minutes
- Stable patients:
 - Refer to these 4 considerations before treating the tachycardia as SVT:
 - HR > 180/min in a child, or >220/min in an infant
 - R-R intervals are regular
 - Width of the QRS at its base must be less than or equal to 1mm (1 small block)
 - Is there an underlying history suggesting a compensatory tachycardia needing fluids?
 - For STABLE SVT – Attempt Vagal Maneuvers
 - If Ineffective:
 - Begin recording EKG strip
 - Administer Adenosine 0.1 mg/kg IV (max of 6mg) with rapid NS flush
 - Repeat Adenosine 0.2 mg/kg IV (max of 12) once

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PEDIATRIC TACHYCARDIA – NARROW WITH A PULSE (Continued)

- Unstable patients (Mottled skin/ Cyanosis, critical hypotension, altered mental status, unresponsive, or ischemic chest discomfort)
 - If the patient is conscious / responsive to pain, give:
 - Versed 0.1 – 0.2 mg/kg IV/IO
 - Consider Versed 0.2 mg IM if no vascular access is available
 - Max dose 2 mg IV or 5 mg IM
 - Synchronized Cardioversion
 - Place monitor into synchronized mode and shock in the following sequence until patient converts:
 - 0.5 Joules/kg, then:
 - 1 Joules/kg, then:
 - 2 Joules/kg (Max dose), repeat at 2 J/kg as needed

PEDIATRIC DYSPNEA – UPPER AIRWAY OBSTRUCTION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of an upper airway obstruction.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Can the patient cough, speak, or breath?
 - Yes – Monitor and transport as indicated
 - No – Apply BLS skills
 - Child older than 1 year
 - 5 back slaps and 5 abdominal thrusts with one hand while supporting the patient from behind
 - Child younger than 1 year
 - Hold baby carefully with head slightly downward and give a series of 5 back slaps and 5 chest thrusts until object is removed
- If patient becomes unresponsive:
 - Begin high quality CPR
 - Look in patients mouth before giving breaths to assess for the obstructing object
 - Do NOT perform blind finger sweeps, only use sweeps to clear a visible object

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO2 reading

PARAMEDIC

- Move to “Needle Cricothyrotomy” protocol as needed
- Paramedics may attempt to visualize the obstruction via laryngoscopy and attempt to remove with forceps while using caution to NOT move the object further into the airway.

PEDIATRIC DYSPNEA – CROUP OR EPIGLOTTITIS (STRIDOR NOTED)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of dyspnea with croup or epiglottitis with stridor present.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Ensure that all measures are taken to keep the patient calm
 - Crying can worsen the inflammation causing the upper airway obstruction
 - You would be within standards of care to forego pain or anxiety inducing procedures in children such as blood pressure assessment, glucose assessment, and IV therapy unless ABSOLUTELY necessary.
- If Stridor/Croup or wheezing without a history of asthma:
 - Administer humidified O2 (3-4 ml NS @ 8 Lpm via Nebulizer)

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO2 reading

PARAMEDIC

- If stridor/croup or wheezing without history of asthma:
 - Administer Nebulized Epinephrine 1 to 2 mg 1:1,000 mixed with 3-4 ml of NS for ages less than 5 years old.
 - Deliver nebulized medication at 6-8 Lpm O2
 - If patient cannot tolerate mask, administer via blow-by
 - In cases of respiratory failure (evidenced by altered mental status or other signs of fatigue) from upper airway obstruction, you can use the nebulizer in-line with a BVM to provide assistive ventilations as needed

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PEDIATRIC DYSPNEA – CROUP OR EPIGLOTTITIS (STRIDOR NOTED) – (Continued)

- If no change in patient condition:
 - Supplement respirations with BVM
 - Intubate as needed
- If allergen exposure – See “Anaphylaxis” protocol
- Respiratory patients should be positioned upright when possible.

PEDIATRIC DYSPNEA – ASTHMA, SUSPECTED / WHEEZING

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of dyspnea with suspected asthma or wheezing.

EMT level may give nebulized Albuterol and Atrovent in the presence of wheezing in respiratory distress patients.

- In order to be able to administer these medications, the EMT must be checked off by their training officer and paperwork must be on file.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
 - Titrate O2 to maintain sats greater than 90%
 - Use BVM to support respirations if respiratory failure is noted
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept
- If patient has a history of asthma, presents with wheezing or poor air movement:
 - Administer Albuterol 2.5 mg in 3 ml NS via nebulizer

AEMT

- If patient is in extremis:
 - Consider administering 0.01 mg/kg epinephrine 1:1,000 IM in thigh.
- Establish IV / Vascular access
 - May administer one albuterol treatment prior to IV access
- Obtain EtCO2 reading

PARAMEDIC

- If patient is still alert:
 - Consider CPAP initiated at 2-5 cmH2O
 - Reference CPAP protocol as needed

CONTINUED ON NEXT PAGE

PEDIATRIC DYSPNEA – ASTHMA, SUSPECTED / WHEEZING (Continued)

- If severe difficulty breathing:
 - Administer 20 mg/kg Magnesium Sulfate (max of 2 grams) mixed in 100-150 ml NS infused over 10 minutes
- Administer 1 mg/kg IV or IM Solu-medrol (Max dose 125 mg)
- Repeat Albuterol treatment after 10 minutes if needed and if IV access has been obtained
 - If no IV access, contact on-line medical control for orders to continue albuterol treatments

PEDIATRIC - SHOCK PROTOCOL (ALL TYPES)

EMR

- Assure CAB's
- Place in supine position as tolerated.
- Pulse Oximetry
- Oxygen via non-rebreather mask.

EMT

AEMT

- Obtain IV / IO access
- Give 20 mL/kg LR bolus (If LR is unavailable, use NS as fluid).
 - May be repeated as needed.
 - Check lung sounds before and after each bolus.
 - Check liver engorgement. It may indicate too much fluid, too fast (swelling to RUQ of abdomen).
 - Use 10 gtt/mL tubing, or other MACRO drip systems for any bolus infusions.

PARAMEDIC

Attempt to determine the etiology of shock by history and exam!!!!Hypovolemic – Hemorrhagic Shock:

- Continue with IV fluid bolus as necessary and titrate to effect to maintain stable perfusion based on the patient's condition (medical or trauma needs), minimally acceptable systolic BP: ($>70 + 2 \times \text{age in years}$).

Anaphylactic Shock:

- Continue with IV fluid bolus and go to the "Anaphylaxis / Allergic Reaction" protocol.

Cardiogenic Shock:

- Go to appropriate protocol.
- After the rate and rhythm normalize, and if the patient is still in shock, then start an Epinephrine (Epi) drip at 0.1-1 mcg/kg/min, carefully titrated to effect. Contact on-line medical control if further orders are needed.

Spinal (Neurogenic) Shock:

- ***If the patient has possible multi-system trauma, withhold vasopressors and consult on-line medical control.***
- Begin Epinephrine drip at 0.1 mcg/kg/min and titrate to effect not to exceed 1 mcg/kg/min.

Septic Shock:

- Initiate IV fluids at 30 mL/kg.
- Move to vasopressors after 30 mL/kg if there is no change. Begin Epinephrine drip at 0.1 mcg/kg/min, carefully titrating to effect, and not to exceed 1 mcg/kg/min without on-line medical control.
- Notify the receiving facility of a possible "Sepsis Alert" patient.

TRANSPORT EMERGENCY

PEDIATRIC ANAPHYLAXIS / ALLERGIC REACTION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of anaphylaxis/allergic reaction.

EMR

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer oxygen via NRB

EMT

- Patients <18 years old, Administer Epinephrine (1:1,000): 0.01 mg/kg IM, maximum each dose 0.3 mg. IM injection is preferred in the THIGH for peds.
- Can repeat once after 15 minutes
- If wheezing/dyspnea is present: Administer Albuterol 2.5 mg / 3 mL via nebulizer, if albuterol is unavailable due to shortages, then a Duoneb can be used.

AEMT

- Obtain IV access
- If hypotensive or inadequate tissue perfusion, Infuse 20 mL/kg of NS.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Apply cardiac monitor
- Use waveform capnography to assess for bronchospasms PRN (shark-fin waveform)
- Administer Antihistamine: H1 Blocker- Diphenhydramine (Benadryl) 1 mg/kg IV/IO max dose of 25 mg. If unable to obtain IV access in pediatrics: Benadryl (1 mg/kg) up to 25 mg deep IM
- Administer Steroid: Corticosteroid – Solu-Medrol (Methylprednisolone) 1 mg/kg IV or IM (maximum of 125 mg)
- If patient is still in extreme Anaphylaxis after treatment above, then consider Epinephrine drip (see Epi drip in medication section) Epi Drip @ 0.1-1 mcg/kg/min, titrate to effect or consult on-line MD for repeat Epi IM doses.
- Signs of extreme / persistent anaphylaxis: profound hypotension (shock), dyspnea, stridor or wheezing.
- Transport as indicated

BRIEF RESOLVED UNEXPLAINED EVENT (BRUE)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of a BRUE.

- An episode in an infant or child less than 2 years old that is frightening to the observer and is characterized by some combination of the following: Apnea (central or obstructive), skin color change: cyanosis, erythema (redness), pallor, plethora (fluid overload), marked change in muscle tone, choking or gagging not associated with feeding, a witnessed foreign body aspiration.
- MOST PATIENTS WILL APPEAR STABLE AND EXHIBIT A NORMAL PHYSICAL EXAM UPON ASSESSMENT BY RESPONDING FIELD PERSONNEL. HOWEVER, THIS EPISODE MAY BE THE SIGN OF UNDERLYING SERIOUS ILLNESS OR INJURY.
- FURTHER EVALUATION BY MEDICAL STAFF IS REQUIRED AND IT IS ESSENTIAL TO TRANSPORT ALL PATIENTS WHO EXPERIENCED AN ACUTE LIFE-THREATENING EVENT (ALTE)

EMR

EMT

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer oxygen as needed
- Be prepared to assist with ventilation if this type of episode occurs again during transport
- Perform initial assessment utilizing the pediatric assessment triangle
- Obtain a description of the event including nature, duration, and severity.
- Obtain a medical history with emphasis on the following conditions: Known chronic disease, evidence of seizure activity, current or recent infections, gastroesophageal reflux, recent trauma, medications (recent or current).
- Assess environment for possible causes.

AEMT

- Obtain IV access as indicated: ONLY if fluids or meds are required.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring is required
- Assess carbon dioxide levels if exposure is suspected
- If the parent or guardian refuses medical care or transport, contact on-line medical control: notify supervisors asap. Involve law enforcement if necessary.

PEDIATRIC BURNS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of burns.

Critical burns, which likely require a burn center (Vanderbilt) would be:

- Burns with > 10% BSA partial thickness involvement or worse in pediatrics.
- Any burns that involve the airway or thoracic region (would affect breathing)
- Burns affecting the genitalia “1%”

Keep patient warm, hypothermia is a complication of critical burned patients.

Focus to prevent infection, use dry sterile dressings (burn sheets) if critical

EMR**EMT**

- Remove burned or smoldering clothes
- Cool with cool (not cold), moist, sterile towels if available
- Burns involving more than 10 percent body surface area should be covered with a dry sterile dressing, preserve heat loss when possible.
- Remove dry chemicals by brushing off the substance, and remove liquid chemicals by flushing with large amounts of water unless contraindicated according to the ERG handbook.
- Assess / Stabilize ABC's as needed
- Oxygen via NRB and control airway as indicated

AEMT

- Obtain immediate vascular access: is the patient hypotensive less than $(70 + (2 \times \text{age}))$ in peds. If yes, initiate Lactated Ringer bolus of 20 mL/kg in pediatrics

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- See “Pain Management” protocol as needed; fentanyl may be given 1 mcg/kg (use caution if patient is hypotensive)
- Be careful using the “broselow” tape, it will give RSI dosing for fentanyl.
- Transport as indicated

PEDIATRIC - DIABETIC EMERGENCY / HYPOGLYCEMIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of a diabetic emergency / hypoglycemia.

EMR

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed

EMT

- If BGL >70 transport as indicated
- Determine if the BGL is greater than 50 mg/dl, If the patient is awake, alert, and cooperative but is **symptomatic** with a glucose level of 50-70 mg/dl, you may give **Oral Glucose** 15 grams (1 tube) PO (oral), buccal, or sublingual. Infants who are hypoglycemic, apply a small amount of oral glucose to a pacifier and allow the infant to consume.
- *** DO NOT GIVE ANYTHING BY MOUTH TO A PATIENT WHO HAS ALTERED MENTAL STATUS***

AEMT

- If the patient is presenting altered / unresponsive with a BGL < 70, establish IV access.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitor
- Paramedic should be the attending provider with pediatric patients receiving medications
- Administer dextrose per the following guidelines: D50 1-2 mL/kg for patients >8 years old
- D25 2-4 mL/kg for patients 6 months to 8 years of age
- D10 2-4 mL/kg for patients from neonate to 6 months of age
- Max rate of infusion 2 mL/kg/min
- If D25 or D10 are not available, utilize a syringe of D50: to make D25, expel 25 mL of D50 and draw up 25 mL of Normal Saline.
- To make D10, expel 40 mL of D50 and draw up 40 mL of NS.
- Reminder: IO is appropriate after 2 failed IV attempts or 90 seconds
- Transport as indicated
- If unable to give dextrose and the patient's mental status is abnormal, then transport emergency

PEDIATRIC HYPERTHERMIA / HEAT RELATED ILLNESS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of hyperthermia / heat related illness

EMR**EMT**

- Immediate cooling has been proven to be more beneficial if done prior to transport from sporting events. If athletic trainers have cooling capabilities, allow them to cool the patient prior to transport.
- Assess / Stabilize ABC's as needed
- Assess temperature
- Administer O2 as needed
- If history is suggestive of heat exhaustion or heat stroke: Remove to cooler environment, cool with moist sheets slowly so that the patient will not start to shiver.

AEMT

- Obtain IV access and administer Lactated Ringers (LR) <60 days: 10 mL/kg, >60 days: 20 mL/kg

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- If seizures are present, then go to the seizure protocol.
- Transport as indicated

PEDIATRIC HYPOTHERMIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of hypothermia.

Keep the doors of the ambulance CLOSED with the heat on high, especially in the winter, 3 layers of blankets are recommended for re-warming, with at least one blanket under the patient between them and spine boards, etc.

Note that with hypothermia as subtle as 96 degrees. The body can lose the ability to form clots / slow bleeding in trauma patients.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Assess temperature
- Remove all wet clothing
- Protect against heat loss and wind chill
- Avoid rough and excessive movement
- Administer oxygen and begin external warming
- If no pulse or breathing: Start CPR. Exception: defibrillate one time and then NO MORE.

AEMT

- Obtain IV access with NS from fluid warmer: <60 days: 10 mL/kg >60 days: 20 mL/kg
- If narcotic ingestion is possible, give Narcan 0.4-4 mg IV

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- Monitor cardiac rhythm: treat only life-threatening arrhythmias
- If no pulse or breathing: Resuscitate per PALS protocol with the following exceptions: Defibrillate 1 time and the NO MORE, Lidocaine may or may not work.
- In pulseless v-tach, v-fib with hypothermia: administer magnesium sulfate 20 mg/kg IV/IO over 10 minutes
- Transport as indicated

PEDIATRIC NAUSEA / VOMITING

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of presentation of nausea / vomiting.

AEMT level may give Zofran IV in pediatric nausea / vomiting situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR**EMT**

- Assess / Stabilize ABC's as needed.
- Control airway and be prepared to suction.
- Attach pulse oximetry.
- Give oxygen as indicated.

AEMT

- Obtain IV access, administer fluids as needed for dehydration. 20 mL/kg LR (Normal Saline may be used as an alternative if LR is unavailable) bolus, reassess and repeat and repeat as needed to restore normotensive blood pressure.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac Monitor, a patient who has been experiencing emesis may be hypovolemic and have electrolyte depletion as well.
- Acquire and transmit a 12 lead ECG if you suspect any cardiac related illness.
- Administer Zofran: >40 kg: 4 mg IV (maximum of 4 mg) <40 kg: 0.1 mg/kg IV (maximum of 4 mg)
 - Transport as indicated

PEDIATRIC OVERDOSE-GENERAL / MEDICATIONS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of an overdose.

**When in doubt, call on-line medical control or TN Poison Control Hotline: 1-(800)-222-1222 or 1-(888)-222-1222
<https://www.vumc.org/poisoncenter/>**

EMR

- Assess / Stabilize ABC's as needed
- Suction as needed
- Oxygen via NRB
- Pulse oximetry
- Aggressive airway control with ventilation if needed
- Obtain history: type and amount of poison, if possible, bring the container with the patient, route of intake, time of intake, history of drug or alcohol usage.

EMT

- Check blood glucose, if <70 or >400, go to the appropriate hypo/hyperglycemia protocol.

AEMT

- Obtain IV access, any hypotension, then give a fluid bolus of NS 1 liter.
- If a narcotic opiate overdose is suspected: Pinpoint pupils, Hypotension, decreased respirations. Administer Narcan 0.01 mg/kg IV, IO, IM, or IN not to exceed 2 mg per dose. May repeat once in 5 minutes.
- Narcan should be given IM in the absence of vascular access.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Apply cardiac monitor
- Intubate as needed (giving Narcan early in opioid overdose may prevent the need to intubate the patient)
- If patient is seizing refer to seizure protocol
- If beta blocker overdose is suspected and the patient is bradycardic and/or hypotensive: Give glucagon 1 mg IVP
- If tricyclic overdose is suspected (contact medical control is not sure of the drug): give sodium bicarb 4.2% 1 mEq/kg IV/IO

CONTINUED ON NEXT PAGE

PEDIATRIC OVERDOSE-GENERAL / MEDICATIONS (Continued)

- If calcium channel blocker overdose is suspected and the patient is bradycardic and/or hypotensive: give calcium chloride 20 mg/kg mixed in a 100 mL bag of NS and given IV/IO over 10 minutes
- If the patient is agitated and a possible stimulant overdose is suspected: Consider versed 0.1 mg IV/IO (max 2 mg IV/IO) or 0.2 mg/kg IM (max 5 mg IM). Repeat dose in 10 minutes and titrate to effect.
- Transport as indicated

PEDIATRIC PERMISSIVE HYPOTENSION / TRAUMA FLUID RESUSCITATION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with hypovolemia / signs and symptoms of shock due to trauma.

EMR

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as indicated

EMT**AEMT**

- Obtain IV access
- Identify normal vital signs, NOT tachycardic: infant 100-180, toddler 100-140, school age 75-120. Normotensive systolic BP ($70 + (2 \times \text{age in years})$). Reassess every 5 minutes with high MOI patients, establish IV access PRN to be ready to treat if S/S of shock develop. **DO NOT GIVE FLUID BOLUS UNLESS S/S OF SHOCK ARE PRESENT.** You may see tachycardia before hypotension.
- Signs of shock: Tachycardia, hypotension, ETCO2 <30 mmHg
- Administer fluid bolus to maintain systolic pressure ($70 + (2 \times \text{age in years})$) or maintain peripheral pulses. <60 days: 10 mL/kg >60 days: 20 mL/kg
- Ensure IV fluids are warm when infusion into trauma patients
- You may consider use of IO device unless area of injury prohibits IO placement when you are unable to obtain peripheral IVs. Fluid administration and rate are the same with IV/IO
- Lactated ringers (LR) is the preferred IV fluid of choice for all patients receiving bolus of isotonic crystalloids.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring required
- Monitor ETCO2
- While fluids are being administered, watch for ETCO2 to increase first as initial sign of improvement, then assess for BP to increase. **DO NOT CONTINUE FLUIDS** once you reach goal systolic BP. You may repeat PRN to maintain target systolic BP.
- Humeral head IO access is **NOT** permitted for pediatric patients less than 8 years old, unless authorized by on-line medical control.

PEDIATRIC POISONING / CHEMICAL EXPOSURE / HAZ-MAT / NERVE AGENTS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of poisoning, chemical exposure, Haz-mat, nerve agents.

EMR

- Personnel safety is the highest priority. Do not handle the patient unless they have been decontaminated. All EMS treatment should occur in the "Support Zone" after decontamination
- Assess / Stabilize ABC's as needed
- Aggressive airway control with ventilation if needed
- Suction as needed
- Administer oxygen via NRB
- Obtain history: Type and amount of poison, if possible, bring the container with the patient, route of intake, time of intake, history of drug or alcohol usage
- If inhaled poison, remove patient from the source using appropriate PPE/SCBA preferred. Assess CO levels if exposure to CO is suspected. Consult with / use Haz-Mat personnel when appropriate.
- If the chemical is a dry substance, then brush off the chemical before irrigating, irrigate with copious amounts of water and reassess for hypothermia.

EMT

- Check blood glucose if <70 or >400, go to the appropriate hypo/hyperglycemia protocol

AEMT

- Obtain IV access: Any hypotension, then give 20 mL/kg in peds.

PARAMEDIC

- Cardiac monitoring as indicated
- Intubate as needed
- If patient is seizing, then go to the seizure protocol.
- If patient is agitated and possible stimulant overdose is suspected: Consider versed 0.1 mg/kg IV/IO (max 2 mg IV/IO) or 0.2 mg/kg IM (Max 5 mg IM) Repeat dose if needed in 10 minutes and titrate to effect.
- For organophosphate / nerve agent poisoning: administer atropine 0.05 mg/kg IVP every 5-15 minutes to dry secretions
- Depending on S/S administer Nerve agent antidote kit: Mild: increased secretions, pinpoint pupils, general weakness: Decontamination, supportive care. Moderate: mild symptoms and respiratory distress, 1 nerve agent antidote kit, may be repeated in 5 minutes, PRN. Severe: Unconscious, convulsions, apnea, 3 nerve agent antidote kits.

PEDIATRIC-SEIZURES

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of pediatric patients with a chief complaint / signs and symptoms of a seizure.

Special attention should be applied to ensure that pediatric patients with seizures have an effective work of breathing. "Disordered control of breathing" can occur easily due to prolonged or acute seizure activity. If you suspect ineffective breathing due to seizures in a child or infant, provide assisted ventilations with a BVM. Use ETCO₂ monitoring to assess for signs of hypercapnia, indicating hypoventilations. (ETCO₂ >45)

EMR

- Assess / Stabilize ABC's as needed
- Protect patient from injury
- Suction as needed
- Nasal Airway (NPA) as needed
- Give O₂ and assist ventilations as needed
- Secondary: Assess vital signs ASAP, pulse oximetry, assess temperature if indicated.

EMT

- Assess blood glucose level

AEMT

- Obtain IV access as indicated

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- Capnography: required if giving versed
- If IV is established and the patient is seizing: Give versed 0.1 mg/kg IVP max dose 2 mg/IV/IO
- If no IV is established and the patient is seizing: Give versed 0.2 mg/kg IM, max dose 5 mg IM.
- If the patient continues to have seizures: IV route-versed 0.1 mg/kg may be repeated 2-3 minutes after the initial dose. Versed IVP may be repeated 2 times, after initial IV or IM dose.
- IM route-versed 0.2 mg/kg IM may be repeated 5 minutes after the initial dose. Versed IM may only be repeated ONE TIME.
- Transport as indicated

LATERAL PATELLAR REDUCTION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of a patellar dislocation.

Assessment/Indications

- Severe Knee Pain
- Lateral displacement of the patella
- Inability to straighten extremity
- Must be ≥ 10 years of age

EMR

EMT

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Pulse oximetry
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading

PARAMEDIC

- Cardiac monitor
- Fentanyl 1 mcg/kg IV to a maximum of 100 mcg.
- Explain the procedure to the patient.
- Refer to next page for the procedure to reduce the patella.

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LATERAL PATELLAR REDUCTION – PROCEDURE (CONTINUED)

PARAMEDIC

1. Stand on the lateral side of the leg on which the patellar reduction is to be completed.
2. Slightly flex the injured leg at the hip to decrease tension on the quadricep muscles.
3. Extend the knee while applying gentle anteromedially directed force on the lateral patellar edge to lift the patella over the femoral condyle.
4. When the reduction is completed, reassess pulse, motor and sensation distally, then immobilize the knee so that the knee is in full extension.
5. Transport to the ED.

SELECTIVE SPINAL IMMOBILIZATION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

Spinal immobilization should be performed on the basis of mechanism of injury and the patient's symptoms. If the patient has major mechanism of injury (MOI) or MOI cannot be ruled out, then always immobilize.

EMR	EMT	AEMT	PARAMEDIC
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- Is the patient significantly obtunded? Head injury, drugs, or alcohol. Does the patient have neck / midline spine tenderness? Does the patient have neurologic deficit? If no, immobilization is not necessary.
- If yes, evaluate the mechanism of injury.
- High risk of injury, defined as: High speed MVC, Roll-over MVC, intrusion into the patient compartment, death in the vehicle, fall 2x patient height, fall onto head, auto vs. pedestrian, diving injury, penetrating trauma with motor/sensation deficits.
- Low risk of injury, defined as: low speed MVC, fall from standing position, penetrating trauma without motor/sensation deficits, ambulatory at the scene.
- Pediatric patients low risk injury, no board or collar if: fall from standing, fall from bed, MVA in car seat, neurologically normal for age, no apparent serious injury.
- High risk injury MOI: provide full spinal immobilization
- Low risk injury MOI: apply C-collar, transport in the position of comfort.
- WHEN IN DOUBT, IMMOBILIZE.
- Transport as indicated.
- Special consideration in pregnant patients: If spinal immobilization is required, attempt to tilt the patient 15-30 degrees to her left side in order to keep the baby in vitro from compressing the patient's inferior vena cava causing hypotension.
- If the patient refuses immobilization, all risks are to be explained to the patient and documented in the narrative along with a witness' signature.
- Any deviation from this protocol requires contact with on-line medical control.

TREATING CHILDREN FROM METH HOMES

TREATING CHILDREN FROM HOMES WITH METH LABS

There are two main concerns when treating children taken from homes with meth labs - **medical evaluation** and **decontamination**.

MEDICAL EVALUATION

Determining the need for immediate evaluation by a health care provider can be accomplished by assessing whether the child is **asymptomatic** or **symptomatic**.

Asymptomatic

Asymptomatic children do not need to be immediately evaluated by a physician.

If the paramedic or police officer finds a child who is completely asymptomatic, there is no need to take that child to the emergency department.

In cases where medical evaluation is required before the child can be placed in an appropriate social setting, then the emergency department evaluation requires only a history and physical exam.

Subsequent to removal from the home, children can be seen at their pediatrician's office, where issues like neglect may also be addressed.

Symptomatic

For the symptomatic child, medical evaluation should happen immediately and should include a physical exam and history. All laboratory testing should be based on findings from the physical exam and/or other concerning symptoms.

Although there are concerns about the long-term effects of these exposures, there is not an initial evaluation or intervention that can identify or change the long-term outcome.

DECONTAMINATION

If there is residue on the clothes of an asymptomatic child, clothing can be changed prior to transfer. Alternately, the children can be given something disposable to sit on (newspaper, diaper, etc.) and clothes can be changed post-transfer. Regardless of symptom status, children should take a shower at some point following removal, but do not need to be decontaminated at the scene.

TIPS FOR TREATING AND DEALING WITH CHILDREN TAKEN FROM HOMES WITH METH LABS:

- Always take the least traumatic approach.
- Do not take toys or any materials from the house.

Poisons commonly found at meth production sites:

- Solvents
- Ephedrine
- Acids
- Iodine
- Lye
- Phosphorus

The dangers from living in a home that houses a methamphetamine lab can include:

- Injury or death from fire or explosions caused by flammable materials used in the lab.
- Chemical burns
- Poisoning from ingestion or absorption of chemicals
- Upper respiratory symptoms, headaches and rash from exposure to chemicals in the environment
- Long-term effects, such as asthma and neurological problems
- Malnutrition
- Emotional and/or physical neglect

A medical toxicologist at Tennessee Poison Center is always available to answer questions. Call 1-800-222-1222 for help 24/7.

Tennessee Poison Center is a program of Vanderbilt University Medical Center.

**For more information on methamphetamine, please visit
www.TNPoisonCenter.org or www.MethFreeTN.org**

OB/GYN NON-DELIVERY COMPLAINTS AND COMPLICATIONS

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

Assessment questions to consider.

- How many pregnancies (Gravida) & how many births (Para) has the pt had?
- How many weeks (trimester) along is the pt? Estimated Date of Delivery? Multiple births expected?
- Any complications? Hypertension (Pre-Eclampsia), Gestational Diabetes, lower extremity swelling, previous ER visits / admits (current and past pregnancies) any previous birth complications?
- Any vaginal bleeding or discharge? How much? How long has it been going on?
- Any pre-natal care, OB visits, what clinic?
- Any trauma prior to onset of symptoms / complaints?

Placental Abruption (Abruptio Placentae)

- **Vaginal Bleeding:**
 - **Nature:** Often **dark red** and may be concealed (hidden hemorrhage) beneath the placenta.
 - **Pain Association:** **Typically, painful.** The bleeding is usually accompanied by abdominal pain.
- **Uterine Symptoms:**
 - **Uterine Tenderness:** The uterus may feel **firm, rigid, or "board-like"** due to muscle contractions.
 - **Contractions:** Frequent and painful uterine contractions are common.
- **Fetal Distress:**
 - **Possible:** Due to compromised placental function, signs of fetal distress (e.g., abnormal fetal heart rate) may be present.
- **Associated Risk Factors:**
 - **Hypertension:** Chronic or pregnancy-induced hypertension.
 - **Trauma:** Abdominal injury or trauma.
 - **Substance Use:** Particularly cocaine use.
 - **Previous Abruption:** History of prior placental abruption

Placenta Previa

- **Vaginal Bleeding:**
 - **Nature:** **Painless and bright red** bleeding.
 - **Timing:** Typically occurs in the **second or third trimester**, often without any abdominal pain.
- **Uterine Symptoms:**
 - **Uterine Tone:** The uterus is usually **soft and non-tender**, without the rigidity seen in abruption.
 - **Contractions:** Generally **absent** unless there is another underlying issue.
- **Fetal Condition:**
 - **Less Immediate Fetal Distress:** Unless bleeding is severe, fetal distress is less common compared to abruption.

CONTINUED ON NEXT PAGE

OB/GYN NON-DELIVERY COMPLAINTS AND COMPLICATIONS (Continued)

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

AEMT level may give Zofran IV, IM or PO in nausea/vomiting situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

EMT

- Place in position of comfort. (**3rd trimester: If supine, place in left lateral recumbent. If spinal precautions are used, tilt board, and pad voids.**)
- Look, or have pt feel for protruding limbs, placenta, fluids.
- Control bleeding as appropriate. Do not insert bandaging into vagina.
- Obtain Vitals including blood pressure, heart rate, breathing rate, oxygen saturation.
- Apply oxygen and airway control as appropriate for condition and your licensure level.

AEMT

- If required, initiate IV saline lock or normal saline / LR @ TKO. If signs of shock are present, then 20ml/kg fluid bolus. May also see **Shock** protocol.
- Consider Glucose check.

PARAMEDIC

- If nauseous, may administer Zofran IV or ODT as appropriate. See Zofran in formulary for contraindications, considerations, and dosing.
- 3-lead monitoring and obtain 12-lead as appropriate.
 - Pregnancy increases fluid volume and strain on the heart. Always consider cardiac monitoring.

Remember to treat the mother's complaints and complications. Always call for orders if unsure.

ACTIVE OR IMMINENT DELIVERY

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol at or below their appropriate level.

Assessment Considerations

- **Term of pregnancy** in weeks, and **original** estimated delivery date?
- Obtain history including number of pregnancies (Gravida) and previous births (Para).
- Any complications with current or previous pregnancy / birth?
- Any pre-natal care or problems?
- Any lower extremity swelling?
- Any other pertinent information?

Important Notes

- The greatest risk to the newborn is airway obstruction and hypothermia. Keep newborn dry, warm, and airway open (if needed; bulb suction mouth then nose, squeeze bulb before insertion.) Always assume respiratory and use positive pressure ventilation if needed.
- The greatest immediate risk to mother is post-partum hemorrhage. Watch for excessive blood loss and hypovolemic shock.
- Consider 2nd unit if either mother or newborn are critical, otherwise keep them together.

EMR**EMT**

- Prepare ambulance with OB kit and keep patient area warm. Prepare mother by removing all lower clothing and preserving dignity as much as possible (gown, blanket, or sheet).
- **Monitor vitals including blood pressure, heart rate, and pulse ox. Apply oxygen and airway support as appropriate.**
- Visually check for crowning, limbs (breech), or placenta, repeat as needed. Do not perform internal examination.
- Do not delay transport. Alert receiving facility as early as possible. Transport as indicated. Be aware of signs of imminent delivery, i.e. contractions less than or around 3-5 minutes apart and 30-60 seconds long.
- If head is crowning, prepare to catch, apply gently pressure to prevent an “explosive” delivery. Do not apply force, push / pull newborn.
- As head delivers, suction mouth then nose with bulb suction. Check for nuchal cord around neck, remove or relieve cord gently.
- After delivery, keep infant on same level as mother, **Dry, Warm, Stimulate** infant. If baby is crying, has good muscle tone, heart rate above 100 bpm, and breathing spontaneously place on mother’s chest. **Keep infant warm and covered.** Check APGAR score @ 1 and 5 minutes apart.
- Clamp cord if no longer pulsing or if infant requires resuscitation. Clamp cord at roughly 8 and 10 inches from baby. Cut in between clamps. Re-assess for bleeding and apply additional clamps as needed.
- Re-evaluate for bleeding, signs of shock, additional births, placental delivery. Maintain ABCs for both.

CONTINUED ON NEXT PAGE

ACTIVE OR IMMINENT DELIVERY (Continued)

AEMT

- Initiate an IV (on mother) with LR @ TKO. Increase rate to 20 ml/kg if signs of shock, maintain blood pressure.
- Check BGL heel stick on newborn. Normal BGL at or above 50.

PARAMEDIC

- Apply 3-lead monitoring to mother.
- **Refer to “Neonate Resuscitation” Protocol as needed.**

ABNORMAL DELIVERY: Breech or Limb Presentation

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

EMR

EMT

- **Breech Presentation**

- Provide oxygen and airway maintenance as appropriate.
- Allow delivery to proceed spontaneously – Do Not Pull.
- Support infant's body as it delivers.
- If head delivers spontaneously then proceed with "Normal Delivery" protocol.
- If head **does not** deliver within **3 minutes**, insert gloved hand into vagina to create an airway for the infant.
- **DO NOT** remove your hand until relieved by Higher Medical Authority.

- **Limb Presentation**

- Position mother supine, with head lowered and hips elevated.
- Transport immediately, do not delay.

AEMT

- Initiate IV TKO, if hypotensive 20 ml/kg.

PARAMEDIC

- Monitor for arrhythmia's
- Contact Medical Control and Transport Emergently

ABNORMAL DELIVERY: AMNIOTIC SAC PRESENTATION & NUCHAL CORD

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

AMNIOTIC SAC PRESENTATION**EMR****EMT**

- Oxygen and airway maintenance appropriate to the patient's condition.
- Place patient in position of comfort.
- Pulse oximetry.
- Amniotic sac:
 - If no fetus visible, cover presenting part with moist, sterile dressing.
- Contact Medical Control ASAP.

AEMT

- Initiate IV NS / LR @ TKO. If hypotensive 20 ml/kg.

PARAMEDIC

- Monitor for arrhythmias

NUCHAL CORD (cord around neck)**EMR****EMT****AEMT****PARAMEDIC**

- Carefully remove cord from around baby's neck by slipping cord over the baby's head if possible.
- Attempt to prevent creating a knot in the cord.
- Prevent cord from strangulating the neonate during delivery.
- Transport as indicated, do not delay.

ABNORMAL DELIVERY COMPLICATIONS:Meconium Staining / Prolapsed Cord

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

Meconium Staining

EMR	EMT	AEMT
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- Do not stimulate breathing till you have bulb suctioned **mouth** then **nose**.
- Administer blow-by oxygen as appropriate for patient's condition.
- Obtain APGAR scores.
- Reassess and Repeat treatments.
- Apply Pulse Ox if patient is not improving.

PARAMEDIC

- Apply monitor if not improving.
- Use meconium aspirator if thick secretions causing airway obstruction still present.

Prolapsed Cord

EMR	EMT
-----	-----

- Position mother supine, with hips elevated as high as possible. Use pillows, sheets, blankets, etc. to pad under her hips. Goal is to use gravity to help hold the head off of the cord.
- Instruct mother to "pant" during contractions to help reduce force of contractions.
- Check for pulse in cord.
 - **No pulse in cord:** Insert gloved hand into vaginal, gently apply pressure to infant's head to restore pulse in cord. Cannot move hand after pulse is restored for any reason until instructed by a doctor. Place moist sterile dressing over exposed cord. Transport immediately.
 - **Pulse found in cord:** Cover exposed cord with sterile moist dressing, transport immediately. Reassess regularly to ensure pulse remains intact.
 -

AEMT

- May initiate IV @ TKO if appropriate.

PARAMEDIC

- Monitor for arrhythmias.

APGAR Score Reference Chart

APGAR Test Scoring			
	Score 0	Score 1	Score 2
A ppearance			
	Blue all over	Blue only at extremities	No blue coloration
P ulse	No pulse	<100 beats/min.	>100 beats/min.
G rimace			
	No response to stimulation	Grimace or feeble cry when stimulated	Sneezing, coughing, or pulling away when stimulated
A ctivity			
	No movement	Some movement	Active movement
R espiration	No breathing	Weak, slow, or irregular breathing	Strong cry

Score of 0-3 = Severely depressed, RESUSCITATE!!!

Score of 4-7 = Moderately depressed, STABILIZE

Score of 7-10 = Stable condition, TREAT as indicated

ASSESS A.P.G.A.R. SCORES AT 1 MINUTE AND 5 MINUTES POST PARTUM!

NEONATAL RESUSCITATION

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

EMR

EMT

- Vigorously dry, warm, and stimulate. Clamp and cut cord per delivery protocol. APGAR Assess for tone, respirations, and pulse.
- If respirations are slow, apneic, labored, or persistent central cyanosis.
 - Reposition airway, suction the mouth then nose if secretions are present.
 - Positive pressure ventilation, (BVM no oxygen) for 1 minute.
- Apply pulse ox. Reassess.
- If heart rate less than 100 bpm, do positive pressure ventilation (BVM / Neo-T). Use as little O₂ as possible. Reassess and continue unless HR drops less than 60 bpm or discontinue if pt improves.
- If heart rate less than 60 bpm, start CPR @ 3 compressions to 1 breath with 100% O₂. Reassess, continue if HR stays less than 60 bpm.
- Do not delay transport.

AEMT

- Initiate IV/IO access, bolus NS/LR @ 10 ml/kg if required.
- Obtain blood glucose via heel stick. If less than 50 mg/dl administer D10 @ 2-4 ml/kg bolus and reassess, repeat as needed. Target of 70-150 mg/dl.

PARAMEDIC

- If heart rate less than 100 bpm and not improving with PPV, **consider** advanced airway.
- Place cardiac monitor.
- If heart rate less than 60 bpm **initiate** advance airway.
- If heart rate remains less than 60 bpm despite 3-5 minutes of CPR @ 3:1, then initiate Epinephrine 1:10,000 at 0.02 mg/kg, every 3-5 minutes, till heart rate greater than 60 bpm.
 - Refer to pediatric weight tape to confirm dosages.
- Consider Hypovolemia, Pneumothorax, Hypothermia, Toxins.
- Do not delay transport

PRE-ECLAMPSIA / ECLAMPSIA

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

Signs and Symptoms

- Headache
- Double vision or seeing spots
- Swelling of face, arms, or legs
- Blood Pressure greater than 140/90
- Seizures
- May occur weeks after delivery

EMR**EMT**

- Assess ABC's.
- Place left lateral when after 20 weeks.
- Administer oxygen therapy as needed.

AEMT

- Initiate IV / IO, may consider a 2nd line.
- Obtain blood glucose.

PARAMEDIC**If symptomatic and systolic greater than 140 or diastolic greater than 90**

- Obtain 12-lead and monitor for arrhythmias.
- Administer 4 grams of Magnesium Sulfate in a 100 mL bag of NS, over 20 minutes.
- If seizure activity is noted, may give Versed 2 mg IVP or 5 mg IM.
- If still hypertensive after Magnesium infusion, may give 10 mg of Labetalol slow IVP, repeat at 20 mg every 10 minutes till target BP drops below 140/90 or other contraindication of Labetalol occurs.
- Contact Medical Control.
- Do not delay transport, transport as indicated.

MATERNAL ACLS / CARDIAC ARREST IN PREGNANCY**Management of the Unstable Pregnant Patient:**

- The patient should be placed in a full left lateral decubitus position to relieve aortocaval compression.
- Intravenous access should be established above the diaphragm to ensure that the intravenously administered therapy is not obstructed by the gravid uterus.
- Precipitating factors should be investigated and treated.
- Pregnant patients are at an **increased risk of aspiration**, hypoxemia and pulmonary edema.
- Systemic hypotension can overwhelm compensatory mechanisms, which attempt to maintain uterine blood flow.

**Chest Compressions in Pregnancy:**

- Guidelines for hand placement, rates, and ratios are the same as with non-pregnant adult patients.
- Keep supine, implement manual LUD during compressions, starting at ≈ 12 to 14 weeks of gestational age.
- There is no literature examining the use of mechanical CPR devices in pregnancy, this isn't advised at this time.

Transport / Operations:

- Transport to a facility that can perform a cesarean delivery may be required when indicated (For out-of-hospital cardiac arrest or cardiac arrest that occurs in a hospital not capable of cesarean delivery).
- If available, transport should be directed toward a facility that can perform PMCD, but transport should not be prolonged by >10 minutes to reach an ER with more capabilities. (PMCD = perimortem cesarean delivery).

Electrical Therapy (Defibrillation) Issues During Pregnancy:

- Current defibrillation protocols should be used in the pregnant patient as in the non-pregnant patient.
- The lateral defibrillator pad/paddle should be placed under the breast tissue when possible.

Breathing and Airway Management in Pregnancy:

- Oxygen reserves are lower and metabolic demands are higher; earlier ventilatory support may be necessary.
- Intubation starting with an ETT with a 6.0- to 7.0-mm inner diameter is recommended.
- Upper airway edema and friability (weakening of tissue) occur as a result of hormonal effects and may reduce visualization during laryngoscopy and increase the risk of bleeding.
- Renal blood flow increases, stressing the kidneys, increasing the tendency for pulmonary edema to develop.

Emergency (ACLS) Medications in Pregnancy:

- In the setting of cardiac arrest, no medication should be withheld because of concerns about fetal teratogenicity.
- Drug metabolism is altered in pregnancy. However, there are no current recommendations for altering dosages of routine ACLS drugs. **Use standard drugs and doses when indicated.**